Convex Optimization: Algorithms and Analysis

Set parameters to minimize a cost —

what kind of problems can we model?

what kind of problems can we efficiently solve?

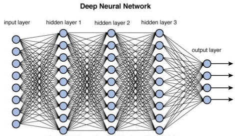
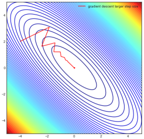
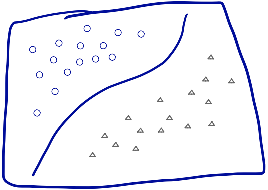


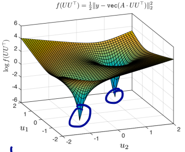
Figure 12.2 Deep network architecture with multiple layers.

Machine Learning & Optimization: a perfect match

very large, important problems, with special structure that we can exploit.







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Research: Decision-making in large scale complex systems....

What this means: Machine Learning, Optimization, Statistical Learning, Large-Scale Networks, Algorithms...

Examples from my group

Robustness to corrupted/missing data

- Systematic errors (not random)
- Recommendation systems with manipulators
- Mixtures of underlying phenomena

Optimization with resource constraints

- Less computation
- Limited memory

Epidemics and (random) processes on graphs

- Diagnosing an epidemic from very noisy data
- Learning a contact graph from epidemics
- Learning graph structure froid noisy measurements

Applications

- Wireless networks, video/image processing, circuit design, air traffic control, ...